

Evaluation and Deployment of 3D-Printed Automatic Weather Stations (3D-PAWS) in Kenya with Integrated Data Visualization and Monitoring Systems

Arthur Kuria¹, Joseph Njoroge¹

¹ Kenya Meteorological Service Authority (KMSA), Kenya

ABSTRACT

This study presents the deployment and evaluation framework of a 3D-Printed Automatic Weather Station (3D-PAWS) in Kenya. It offers a cost-effective and scalable alternative to conventional Automatic Weather Stations (AWS). Reliable meteorological data is crucial for sectors like agriculture, aviation, and climate monitoring. However, standard AWS systems often face challenges such as high costs, complex maintenance, and limited coverage, particularly in Regional Association I (RA I).

The 3D-PAWS system comprises multiple environmental sensors, including those for temperature, humidity, pressure, rainfall, wind, and solar radiation, all housed within a modular 3D-printed structure. The system runs on a solar-battery setup and uses a cellular-enabled microcontroller to send data in real time to cloud-based platforms. Users can access this data through the CHORDS Portal (<https://3d-fewsnet.icdp.ucar.edu>). We are also developing a custom dashboard for the Kenya Meteorological Department to facilitate ongoing monitoring, visualization, and management of a network of stations.

A significant aspect of the system is its local fabrication approach. 3D printing, assembly, and initial sensor calibration occur in-house at the Kenya Meteorological Service Authority (KMSA), Kenya Workshop. This approach improves system adaptability, lowers deployment costs, and promotes sustainable scaling of weather monitoring infrastructure.

This paper outlines the system design, deployment architecture, and data infrastructure of 3D-PAWS in Kenya. We propose a comparative analysis with high-end AWS and manual observations to evaluate performance and reliability, as a foundation for further analysis.

List of Acronyms:

3D-PAWS	3D-Printed Automatic Weather Station
AWS	Automatic Weather Station
KMSA	Kenya Meteorological Service Authority
FEWS NET	Famine Early Warning Systems Network
UCAR	University Corporation for Atmospheric Research
WBGT	Wet Bulb Globe Temperature
ASAL	Arid and Semi-Arid Lands
CHORDS	Cloud-Hosted Real-time Data Services
RA I	Regional Association I

1. INTRODUCTION

Reliable and timely weather data is vital for supporting activities like agriculture, aviation, disaster risk management, and climate monitoring. In Kenya, increasing climate variability and the demand for localized forecasting underscore the need for expanding meteorological observation networks. While traditional Automatic Weather Stations (AWS) are highly reliable, they often entail high acquisition costs, complex maintenance, and limited distribution, especially in remote areas.

Recent progress in low-cost sensing technologies, open-source electronics, and 3D printing has led to new weather monitoring solutions. One notable development is the 3D-Printed Automatic Weather Station (3D-PAWS), which provides a flexible, customizable, and affordable method for collecting environmental data. By using 3D-printed parts and commonly available sensors, 3D-PAWS lowers deployment barriers while maintaining acceptable measurement accuracy.

Simultaneously, cloud-based data platforms and visualization tools have changed how meteorological data is accessed and used. Systems like the CHORDS Portal (<https://3d-fewsnet.icdp.ucar.edu>) allow centralized data storage and real-time monitoring. Custom dashboards enhance management capabilities, data visualization, and performance tracking.

This paper details the design and deployment framework of 3D-PAWS in Kenya, highlighting how it can complement existing meteorological infrastructure. The study focuses on system architecture, local fabrication processes, and data infrastructure, with plans to evaluate performance against high-end AWS and manual observations in future analyses.

The implementation of 3D-PAWS has progressed from pilot programs to an expanding national network. Initial deployments started with prototype stations through collaborative efforts, leading to large-scale installations across various counties. So far, over seventy stations have been deployed through partnerships with national and international organizations, demonstrating the system's scalability. Field experiences from these deployments have provided valuable information about performance, maintenance needs, and operational challenges in varying environmental conditions.

2. SYSTEM DESIGN AND FABRICATION

The 3D-Printed Automatic Weather Station (3D-PAWS) as shown in Figure 1 is designed as a modular and scalable environmental monitoring system. Its physical structure consists of 3D-printed parts, allowing for quick prototyping, easy replication, and adaptation to different environments.

A critical element of the system is its local fabrication approach. Structural components are 3D-printed, and the station's assembly, including sensor integration, wiring, and enclosure fitting, is done in-house. This helps ensure quality control, reduces reliance on outside suppliers, and significantly cuts production and deployment costs.

The system comprises various low-cost sensors to measure key meteorological parameters. Temperature is measured using MCP9808 sensors, while humidity is recorded with SHT31 sensors. Atmospheric pressure is monitored using BMP280 or BME280 sensors. Wind speed and direction are captured by a cup anemometer with Hall-effect sensing and an AS5600-based wind vane. Rainfall is recorded using a tipping bucket rain gauge, and solar radiation levels are measured with SI1145 sensors.

All sensors connect to a Boron microcontroller configured to form part of the datalogger as shown in figure 2, which offers cellular connectivity for real-time data transmission. The setup is powered by a combination of solar energy and battery storage, making it suitable for remote or off-grid areas.

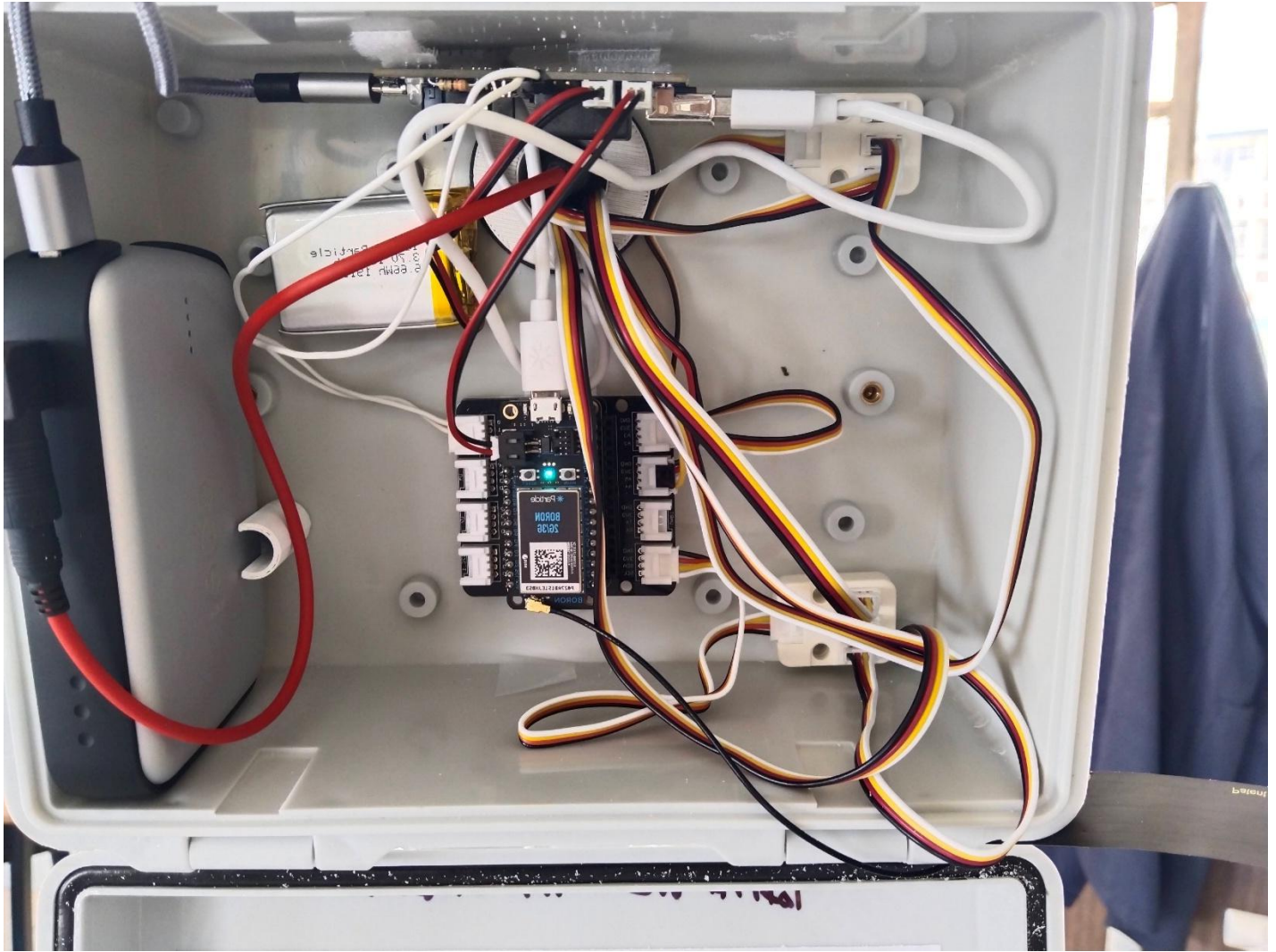
During assembly, we perform preliminary calibration and testing of sensors to ensure acceptable accuracy before deployment. This involves basic validation and alignment of sensor outputs, which improves the consistency and reliability of the data collected. The modular system design allows for straightforward maintenance, replacement of sensors, and future upgrades, making it ideal for wide-scale deployment across various and diverse regions especially in ASAL Areas.

Figure 1 3D-Printed Automatic Weather Station



Time
Health
Battery Voltage
Battery charge status
Cell signal strength
Rain Gauge 1
Rain Gauge 2
Rain Gauge 1 Total Today
Rain Gauge 2 Total Today
Rain Gauge 1 Total Prior
Rain Gauge 2 Total Prior
BMX Temperature 1
BMX Pressure 1
MCP Temperature 1
SHT Temperature
SHT Humidity
SI1145 Visible 1
SI1145 Infrared 1
SI1145 Ultraviolet 1
Wind Speed
Wind Direction
Wind Gust
Wind Gust Direction
Heat Index
Wet Bulb Temperature
Wet Bulb Globe Temperature

Figure 2: Datalogger



3. Field Deployment and Upgrades

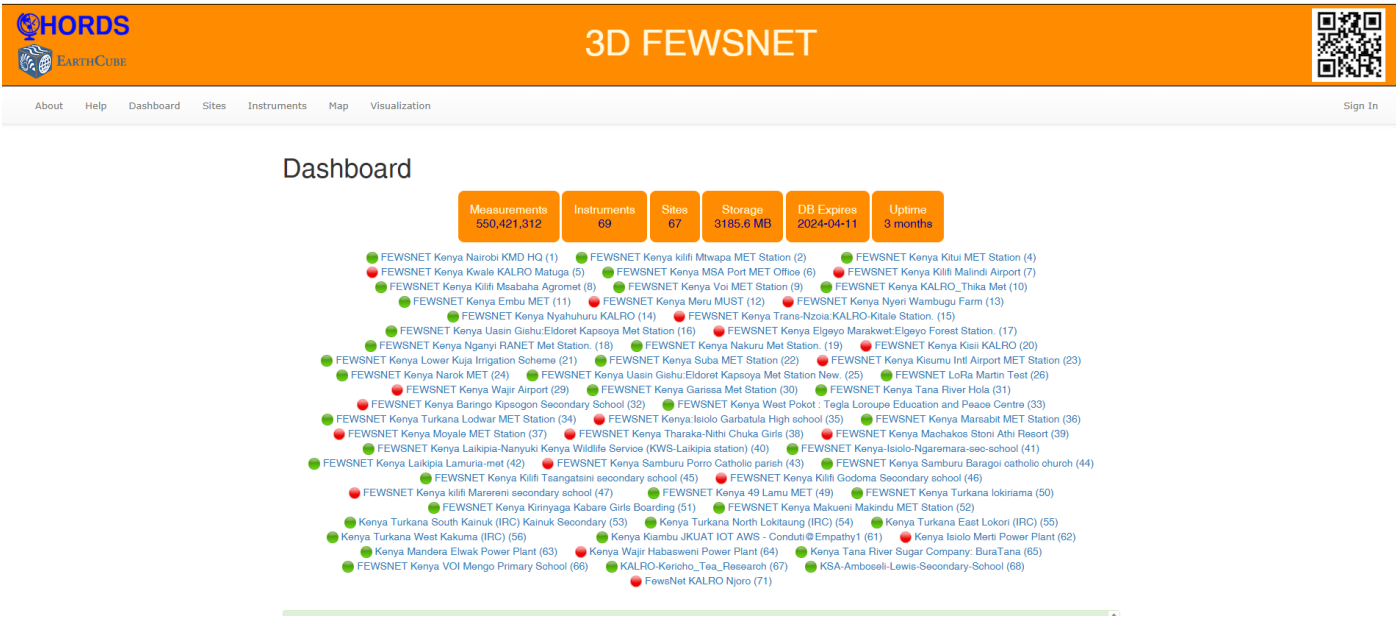
After initial deployment, the 3D-PAWS network underwent upgrades and maintenance to improve performance and expand functionality. Early installations lacked wind and radiation sensors, requiring later upgrades that included adding anemometers and wind vanes to several stations.

These upgrade efforts were coordinated and included inspecting, repairing, and replacing faulty components. The process highlighted the adaptability of the 3D-PAWS system, allowing new sensors to be added to existing stations without needing a complete system overhaul. The deployment strategy also prioritized collaboration with local organizations, such as meteorological departments and research institutions, to ensure security, accessibility, and operational support for the stations.

4. Data infrastructure and visualization

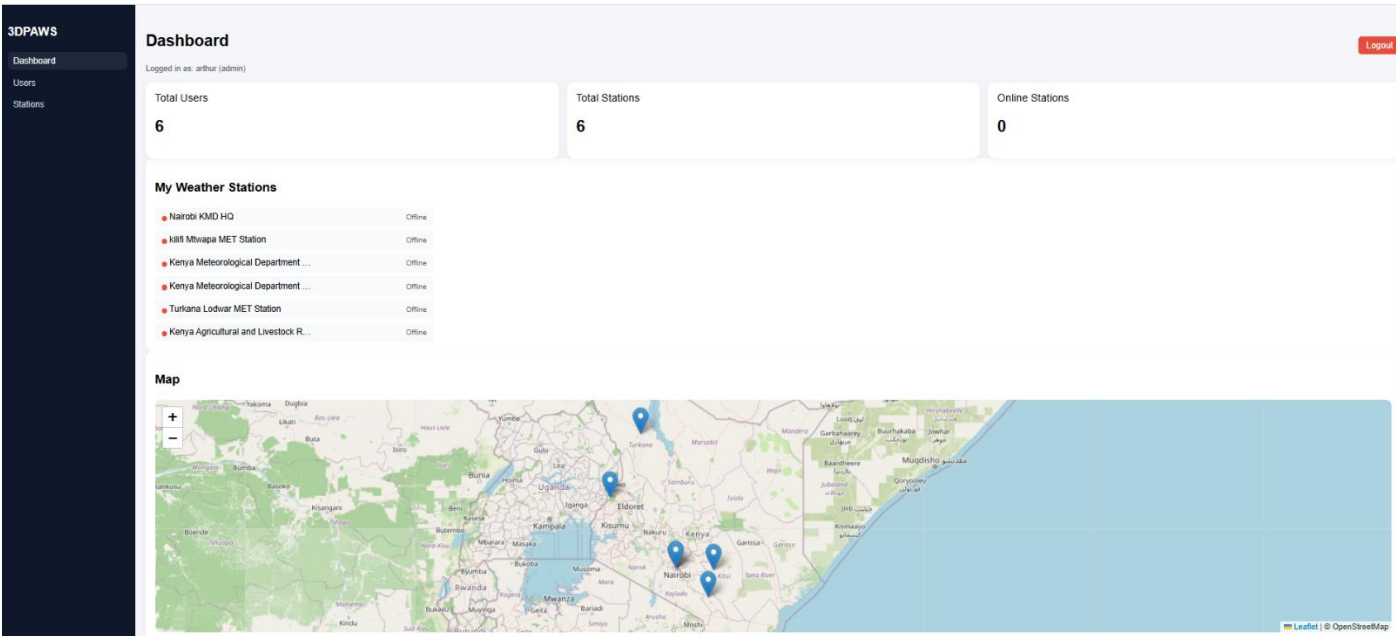
Data collected from the 3D-PAWS network is transmitted via cellular networks to a centralized cloud platform, where it is stored and made available for analysis. The CHORDS platform as shown in figure 3 offers real-time and historical visualizations of meteorological parameters, enabling remote monitoring of multiple stations across different locations.

Figure 3: Chords Dashboard



To improve usability and monitoring, we are designing a custom dashboard as shown in figure 4 to complement the existing data platform. The dashboard will collect large amounts of meteorological data and displays key system metrics, such as the number of active stations, instrument availability, data storage usage, and overall system uptime.

Figure 4: Dashboard custom Dashboard.



The platform will include visual indicators to show the operational status of each station. This will allow for quick identification of faults or data gaps, improving maintenance efficiency and overall system reliability. Integrating cloud-based infrastructure with a custom visualization layer will significantly boost the accessibility and usability of meteorological data. This will support real-time decision-making, system diagnostics, and efficient management of a distributed weather station network.

5. Operational Challenges and Maintenance Considerations

Field observations from deployed 3D-PAWS stations revealed several operational challenges important for evaluation and improvement. The most common issue was the clogging of rain gauges from dust, debris, and organic matter, which affected measurement accuracy. In some cases, internal components like tipping bucket magnets showed rust or mechanical wear.

Environmental factors also posed challenges. High humidity and rainfall in certain areas led to moisture entering data logger enclosures, while insect infestations were seen at several stations, affecting electronic components. Additionally, network connectivity issues were reported in remote regions, hindering real-time data transmission.

Instances of physical damage and vandalism were also noted, emphasizing the need for secure installation sites and community awareness. These challenges highlight the importance of routine maintenance, better enclosure design, and increased protective measures.

Despite these obstacles, most stations were successfully restored to operational status following maintenance and upgrades. This demonstrates the resilience and recoverability of the 3D-PAWS system in field conditions.

To evaluate the effectiveness of the 3DPAWS system under real-world conditions, a comparative methodology was adopted.

6. Methodology

A comparative observational study was performed to assess the performance of the 3D-Printed Automatic Weather Station (3D-PAWS) against high-end Automatic Weather Stations (AWS) managed by KMSA, and manual rainfall observations.

Meteorological data from the 3D-PAWS system was collected through the CHORDS platform, which provides real-time and historical time-series data. Corresponding data from KMSA AWS systems served as benchmarks, while manual rain gauge readings were used as reference measurements for rainfall validation.

The analysis targeted key meteorological parameters, including temperature, humidity, rainfall, wind speed, and wind direction. Data were collected over a consistent observation period, and measurements from the different systems were aligned temporally for a direct comparison.

Time-series analysis was used to evaluate trends, consistency, and differences between the systems. For rainfall, cumulative totals were compared against manual gauge readings to check accuracy, particularly during different rainfall intensities. For temperature and humidity, we analyzed the differences in simultaneous observations to determine how closely they agreed. Wind measurements were assessed based on general trend consistency and variability.

Apart from measurement comparison, we assessed system performance regarding data availability, transmission reliability, and operational stability. Information from the CHORDS platform helped evaluate system uptime, identify data gaps, and gauge the overall robustness of the 3D-PAWS network.

This methodology offers a framework for analyzing both the measurement performance and operational effectiveness of the 3D-PAWS system relative to established meteorological observation methods.

Field observations from maintenance and upgrade activities were also integrated into the analysis to explore operational challenges and system robustness. These observations offered qualitative insights into factors affecting measurement accuracy, system reliability, and maintenance needs in real-world deployment conditions.

References

- 3DPAWS UCAR site
- FEWS NET / CHORDS platform